

King AbdulAziz University
Faculty of Computing and
Information Technology
Computer Science Department

CPCS – 214
Computer Organization and
Architecture
Final Exam

جامعة الملك عبدالعزيز
كلية الحاسبات وتقنية المعلومات
قسم علوم الحاسبات

1st – 1432/1433H

Monday 15/ 2 / 1433H

Time: 4:15-6:15 PM

54
60

Name: Abdullah Bajaber ID: 0857119 Section: CA

15

1. (15 Points) For each of the following multiple choices choose the correct answer:

1.	A certain wafer contains 100 dies 70 of which are perfect; the yield is equal to ____.
☒	a. 0.3 ☒ b. 0.7 c. 30 d. 70
2.	The decimal number equivalent to the following 2's complement number is ____. 1111 1111 1111 1111 1111 1111 1110 0100.
☒	a. -4 b. -27 ☒ c. -28 d. -2,147,483,620
3.	After executing the second instruction (2) with the content of registers as shown, the next instruction that will be executed is instruction ____.
☒	1: sltu \$t0, \$s0, \$s1 2: beq \$t0, \$zero, 4 3: addi \$t0, \$t0, 1 4: addi \$t0, \$t0, 2 \$s0 = 0000 0000 0000 0000 0000 1000 0000 0000 \$s1 = 1111 1111 1111 1111 1111 1111 1111 0000 a. 1 b. 2 ☒ c. 3 d. 4
4.	The CPU time required to execute a program with 12.0×10^9 instructions running on a 2 GHz machine and CPI of 3 is equal to ____ sec.
☒	a. 1.5 b. 6.0 c. 12.0 ☒ d. 18.0
5.	The hazard in executing the following instructions can be solved by ____.
☒	and \$t5, \$t4, \$t1 sub \$t6, \$t2, \$t4 a. prediction b. reordering c. forwarding ☒ d. no hazard

6. ✓	Pipelining improves performance by increasing _____. a. clock rate ☒ b. throughput c. instruction count d. cycles per instruction
7. ✓	The normalized equivalent of the binary number: 0.00111_{two} is: ☒ a. 1.11×2^{-3} b. 1.11×2^3 c. 1.11×10^3 d. 111.0×2^5
8. ✓	_____ technique always updates cache and memory but it makes writes take longer. a. Valid b. Reference c. Write back ☒ d. Write through
9. ✓	Functional units used during jump instruction are all of the following except _____. a. PC b. adder ☒ c. registers d. memory
10. ✓	Upper level memory accessed 20 times with 5 misses; so the hit ratio is equal to _____. a. 0.25 b. 0.5 ☒ c. 0.75 d. 5.0
11. ✓	_____ is a situation that prevents starting the next instruction in the next cycle. a. Stall b. Control ☒ c. Hazard d. Forwarding
12. ✓	_____ means that the items near those accessed recently are likely to be accessed soon. a. Block access ☒ b. Spatial locality c. Temporal locality d. Cache access
13. ✓	During execution of set-less-than (slt) instruction, the PC content is updated _____. ☒ a. 1 time b. 2 times c. 3 times d. undetermined
14. ✓	A need to wait for previous instruction to complete data read/write is _____ type of hazards. ☒ a. data b. control c. structural d. none of the above
15. ✓	_____ is the time required to access the upper level of the memory hierarchy. ☒ a. Hit time b. Access time c. Miss penalty d. Write through

7

2. (10 Points)

Write MIPS instructions to (do not use pseudo-instructions):

a. Decrement register \$t1 by 5.

3 `addi $t1, $t1, -5`b. Load register \$t2 by the 32-bit constant: 0000 0000 0000 1010¹⁰ | 0000 0000 0000 0110⁶.2 `lui $t0, 10`
`ori $t0, $t0, 6``lw $t2, 0($t0)` ✗

c. Multiply register \$s3 by 35.

`addi $t0, $zero, 0` # Loop counter

Loop: `slti $t1, $t0, 34`

2 `beq $t1, $zero, Exit`

`add $s3, $s3, $s3`

`addi $t0, $t0, 1`

Exit: `j _ Loop`

4

3. (5 Points)

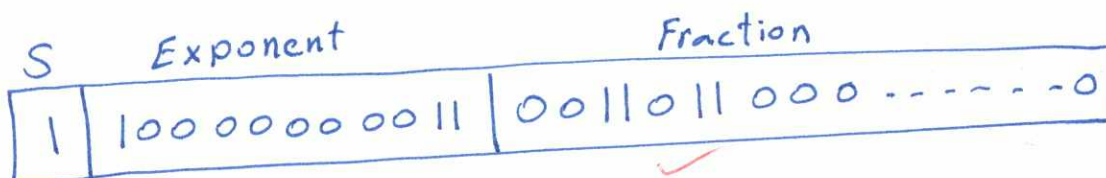
Represent the decimal number -38.75_{10} in the IEEE 754 binary double precision format.1 $S = 1$ ✓

$$38.75_{(10)} = 100110.11_{(2)} = 1.0011011 \times 2^5$$

1 Exponent = $5 + \text{bias} = 5 + 1022 = 1027$

$= 10000000011$

2 Fraction = 0011011000...0

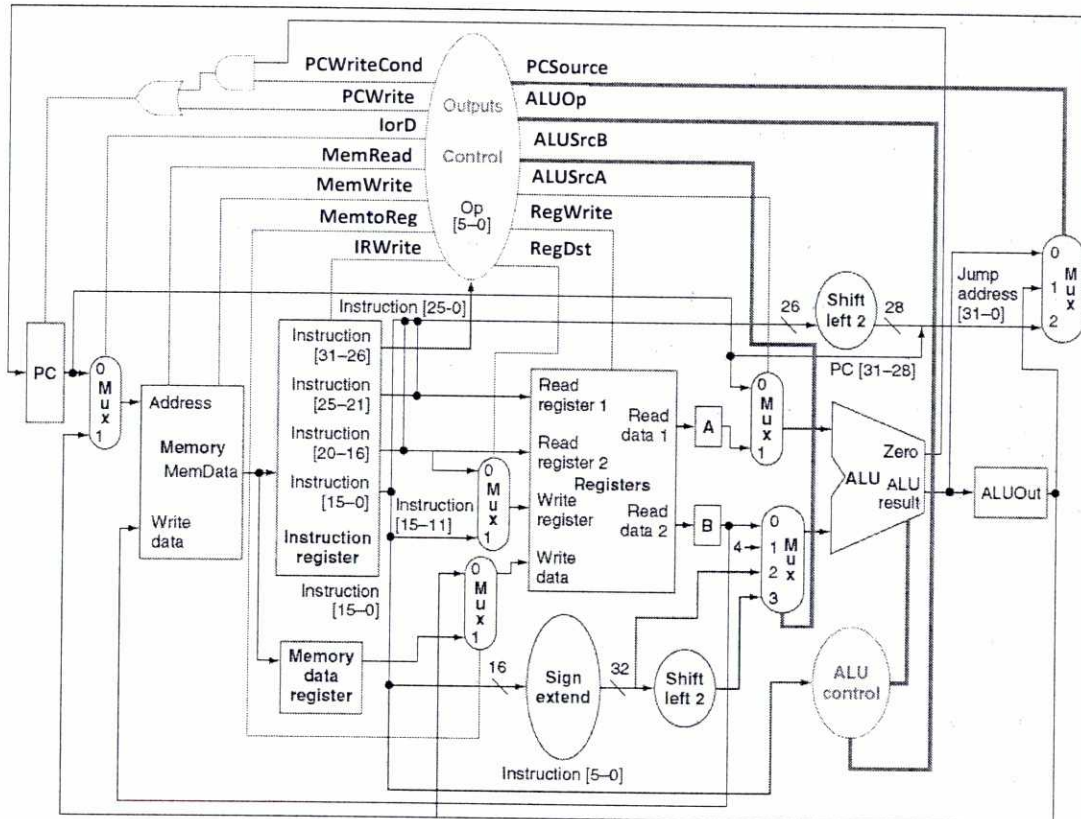


4. (10 Points)

The multi-cycle MIPS architecture used for instructions execution is shown in the figure. During branch instruction completion the following action is performed:

if (A == B) PC <= ALUOut;

ALUOp	value
add	00
subtract	01
funct	10



Illustrate the execution of the branch completion stage by:

a. Listing all of the used functional units.

- Registers: A, B, ALUOut
- ALU
- Control
- ALU control

- PC
- AND gate
- OR gate

b. Listing all of the used control signals and their logical values.

- ALUSrcA = 1
- ALUSrcB = 00
- ALUOp ~~ALU control~~ = 01
- PCWriteCond = 1
- PCSource = 01

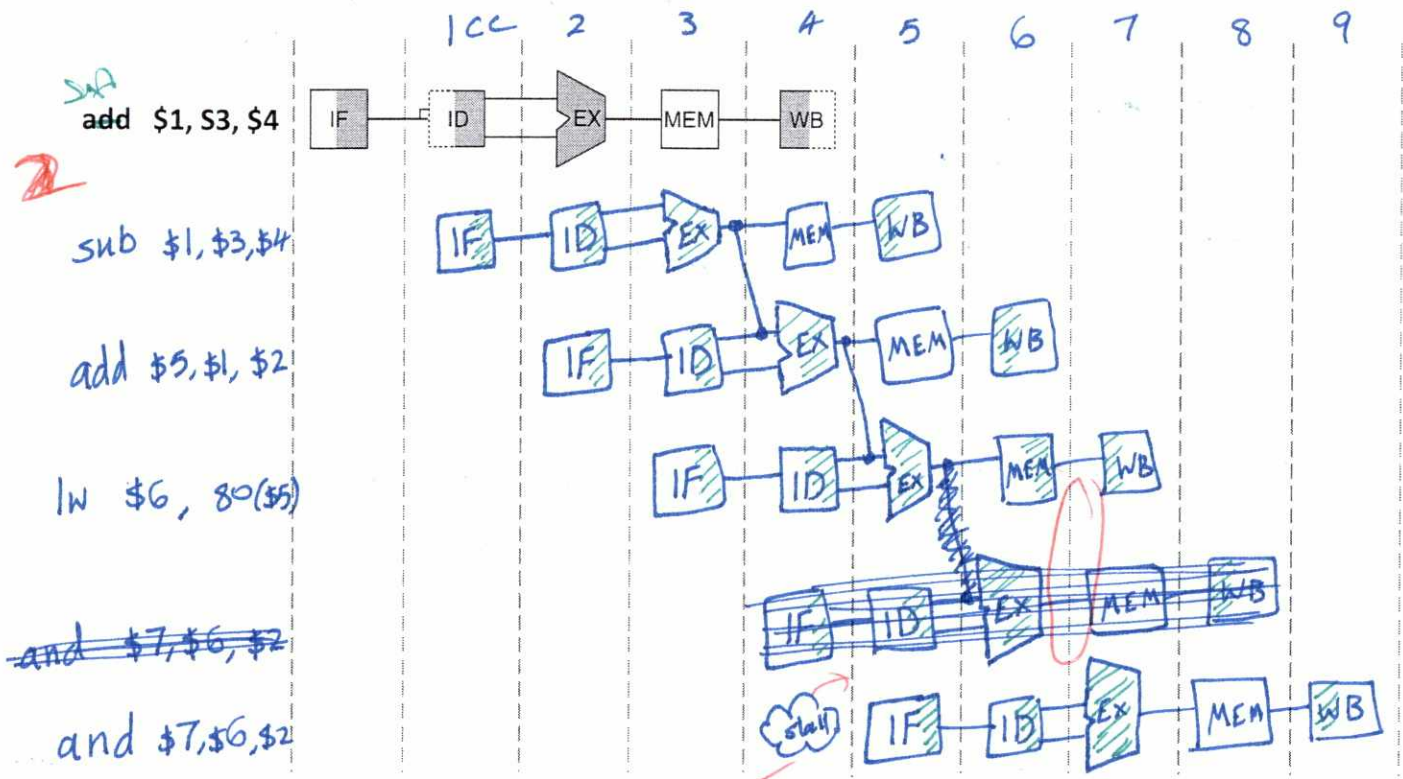
9

5. (10 Points)

A pipelined architecture is used during the execution of the following instructions:

- 1: sub \$1, \$3, \$4
- 2: add \$5, \$1, \$2
- 3: lw \$6, 80(\$5)
- 4: and \$7, \$6, \$2

a. Complete the drawing and show the forwarding paths needed to execute the instructions.



b. How many clock cycles needed using non-pipelined architecture?

$$2 \quad \text{cycles}_{\text{nonpipelined}} = 4 \text{ instructions} \times 5 \frac{\text{cycles}}{\text{instruction}} = 20 \text{ cycles}$$

c. How many clock cycles needed using pipelined architecture?

$$2 \quad \text{cycles}_{\text{pipelined}} = 9 \text{ cycles}$$

d. Can you reduce the number of clock cycles using pipelined architecture? Justify your answer.

3 No. We can't avoid the stall (load-use hazard), because we can't reorder the code due to the dependency of each instruction on the previous instruction.

If the question means how ~~one~~ pipelining ~~can~~ reduces clock cycles, then the answer is "pipelining executes multiple instructions at the same time (in parallel), so we can execute more instructions in a given period of time."

6. (7 Points)

A computer system uses cache memory of size 4 blocks with 1 word per block. The current state of the cache and its contents are shown in the table.

Index	V	Tag	Data
00	Y	000	11011110
01	N		
10	Y	101	01100000
11	N		

a. Show whether request for word address: 8 is a hit or miss.

$$\text{cache address} = \text{block address} \bmod \text{\# of blocks in cache}$$

$$= 8 \bmod 4 = 0 \quad \text{Tag} = 010$$

the tag doesn't match!
Miss!

b. Show whether request for word address: 29 is a hit or miss.

$$29 = \underbrace{1111}_{\text{tag}} \underbrace{01}_{\text{index}}$$

the V bit is "N"

Miss!

c. Compute miss penalty if the system uses 2 words per block, and 1 word wide DRAM organization; it takes 2 bus cycles to send address, 6 bus cycles per DRAM access, and 2 bus cycles to send word of data.

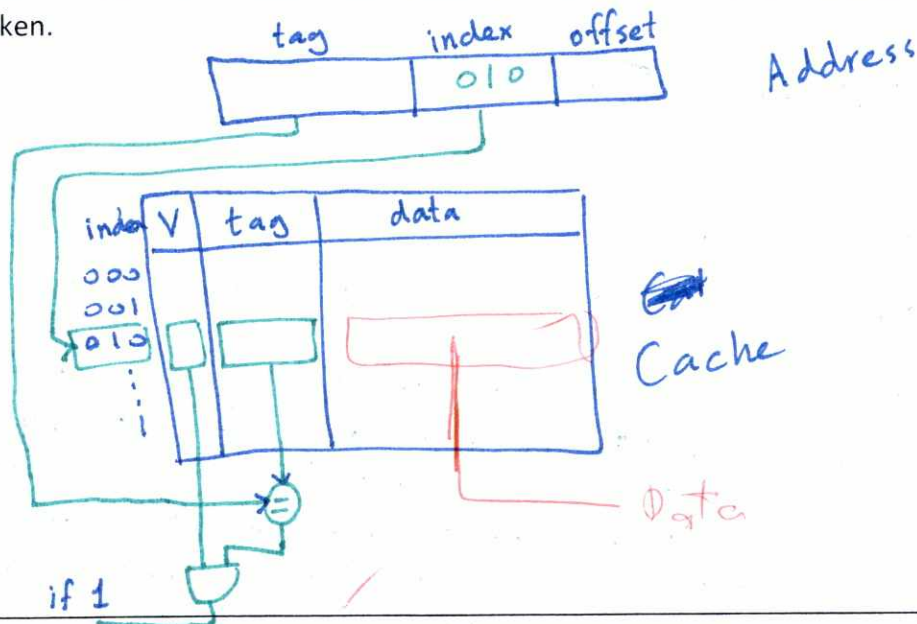
$$\text{Miss penalty} = 2 + 2 \times 6 + 2 \times 2 = 18 \text{ cycles}$$

d. Compute the memory bandwidth.

$$\text{Bandwidth} = \frac{(2 \times 4) \text{ B}}{18 \text{ cycles}} = 2.25 \text{ B/cycle}$$

$$2.25 \sqrt[3]{\frac{18}{16}} = \frac{2.0}{1.6} = 1.25$$

e. Show how a referenced address is divided into a cache index and a tag field and how hit/miss decision is taken.



$2 \frac{1}{2}$ 7. (3 Points)

A 16-bit address computer architecture uses cache memory of size 256 bytes which is divided into blocks each is of size 2 words.

a. What block number does address 1020 map to?

1 word = 4 Bytes

1 block = 2 words = 8 Bytes

$$\text{block address} = \left\lfloor \frac{\text{byte address}}{\# \text{ bytes in a block}} \right\rfloor = \left\lfloor \frac{1020}{8} \right\rfloor = \lfloor 127.5 \rfloor = 127$$

$$\# \text{ of blocks in cache} = \frac{256}{8} = 32 \text{ blocks}$$

$$\text{cache address} = 127 \bmod 32 = 31$$

b. What is the offset of address 1013? (Note: $1023_{\text{ten}} = 111111111_{\text{two}}$).

$1013 = 1111110101$

Byte offset

word offset

= ?

c. How many bits represent the Tag?

